

VESPERA

Mobile Client & In-Vehicle Vision Upgrade

Strategic Investor Briefing — July 2026

EXECUTIVE SUMMARY

Vespera is a mature, production-ready self-hosted AI assistant featuring persistent memory, background cognition, and tool execution. This upgrade introduces a mobile client (PWA + native app) with integrated phone-camera vision capabilities for in-car and on-the-go use. The result is a sovereign, privacy-first AI companion that extends Vespera from desktop/server into the mobile and automotive domains — unlocking new use cases in personal productivity, defence-adjacent situational awareness, and intelligent vehicle assistance.

Key Investment Highlights

- Zero new infrastructure — leverages existing Flask backend & local models
- High-leverage feature: Windscreen-mounted camera + local vision pipeline
- Direct path to in-vehicle AI (CarPlay / Android Auto ready)
- Strengthens sovereign AI positioning for defence and government interest
- Low development cost, rapid MVP (prototype in < 2 weeks)

CURRENT STATE

Vespera already delivers a robust self-hosted experience: persistent multi-layer memory (SQLite nesting-doll architecture), background thinking loop, Telegram bot for phone access, TTS voice responses, web search, financial data, and optional cloud handoff. Users can already reach it via Tailscale or local network from any browser. The missing piece is a purpose-built mobile experience and real-world sensor integration — specifically vision.

PROPOSED UPGRADE

1. Mobile Client Layer

- Progressive Web App (PWA) enhancements for immediate phone/browser access (add-to-home-screen, offline caching, push notifications).
- Native cross-platform app (Flutter recommended) with optimised UI for driving (large touch targets, voice-first mode, CarPlay/Android Auto support).
- Seamless Tailscale or direct API connectivity to existing Vespera backend — no port forwarding required.

2. Vision Tool — Windscreen Camera Integration

The flagship capability: allow the user's phone (mounted on windscreen) to act as a live vision sensor. Frames or processed detections are streamed securely to the Vespera backend for analysis by local vision models (Ollama + Qwen2-VL / LLaVA or lightweight YOLO).

- Real-time road scene understanding (vehicles, pedestrians, signs, hazards, construction).
- Contextual narration & alerts via TTS — hands-free, eyes-on-road assistance.
- Automatic memory ingestion: Vespera extracts facts from drives ("Construction on Gateway Motorway at 17:40").
- On-demand or motion-triggered capture for later review or agent reasoning.
- Optional integration with SEL quad-brain architecture (route vision data to Executor / Strategist / Critic agents).

TECHNICAL APPROACH

Component	Technology	Notes
Frontend	Flutter (preferred) or React Native + VisionCamera	High performance camera, CarPlay support, single codebase
Backend Extension	New /vision endpoint in existing Flask api.py	Image upload, local inference, structured JSON return
Vision Pipeline	Ollama + vision model or YOLO + OpenCV	Fully local; optional on-device fallback (TFLite)
Connectivity	Tailscale + HTTPS / WebSockets	Zero-config remote access; encrypted
Privacy & Security	TLS, API tokens, local-first processing, face/plate blurring option	Sovereign by design; user-controlled

STRATEGIC VALUE & MARKET OPPORTUNITY

This upgrade transforms Vespera from a powerful desktop AI into a true mobile sovereign companion. In-vehicle vision is a high-visibility, high-utility feature that resonates with both consumer and enterprise/defence audiences.

Use Cases

- **Personal / Commuter:** Intelligent co-pilot that remembers routes, warns of hazards, logs trips automatically.
- **Defence / Sovereign:** Edge situational awareness in GPS/comm-denied environments; ROE-compliant analysis; data stays on Australian hardware.
- **Fleet / Enterprise:** Privacy-first driver assistance and event logging without cloud telemetry.
- **Research / SEL Evolution:** Real-world sensor data feeds the multi-agent system, accelerating emergent properties and “digital life” experiments.

Competitive differentiation: Most AI assistants are cloud-tethered. Vespera + mobile vision delivers private, always-available, context-rich intelligence that competitors cannot match without sacrificing sovereignty.

IMPLEMENTATION ROADMAP

Phase	Deliverable	Timeline	Effort
MVP	PWA polish + basic camera capture/upload + simple vision endpoint	1–2 weeks	Low
Core Vision	Native Flutter app, motion-gated streaming, TTS alerts, memory integration	3–5 weeks	Medium
Advanced	CarPlay/Android Auto, SEL agent routing, on-device fallback, event search UI	6–10 weeks	Medium-High

RISKS & MITIGATION

- **Privacy / Legal (dashcam laws):** Fully local processing, user consent gates, optional blurring. Complies with Australian privacy expectations.
- **Performance on mobile:** Intelligent frame selection + compression; optional on-device inference fallback.
- **Driver distraction:** Voice-first design, minimal visual UI while driving, CarPlay integration.
- **Model accuracy:** Start with proven open models; continuous improvement via memory feedback loop.

NEXT STEPS & ASK

We are ready to move immediately. The MVP can be demonstrated within two weeks. Full vision pipeline and native app follow rapidly. This upgrade materially increases Vespera's addressable market, technological moat, and appeal to sovereign/defence stakeholders while remaining true to the core local-first, no-subscription philosophy.

Recommended immediate action: Approve prototype budget for mobile client + vision tool. Target: working demo by end of month.

This document is confidential and intended for prospective investors and strategic partners only. All technical claims are based on the current Vespera codebase and proven open-source components.